

# Ricardo Manuel da Encarnação dos Santos

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## Personal Summary

**Senior Software Engineer** with deep expertise in C#, ASP.NET Core, and Python, specializing in cross-platform backend services, distributed systems, and real-time telephony APIs.

**Production AI/ML Experience:** Track record of architecting scalable, multi-tenant Kubernetes services to train, deploy, and integrate NLP/deep learning models into live enterprise systems.

**End-to-End Ownership:** Complete lifecycle expertise spanning system architecture, observability (OpenTelemetry, Prometheus, Grafana), CI/CD pipelines, and cloud/on-prem deployments.

## Professional Experience

### Enghouse Systems

#### Senior Software Engineer

Feb 2022 - Present | Lisbon, Portugal (Remote, Full-time)



- **Served as lead developer** for a cross-platform UCaaS backend service, guiding the technical implementation of a unified API that bridges multiple telephony platforms across Linux, Windows, Kubernetes, and on-premises environments.
- **Engineered robust server-side applications** using C#, ASP.NET Core, Entity Framework, PostgreSQL, and Python within a geographically distributed engineering team.
- **Integrated enterprise observability** by embedding OpenTelemetry collectors to export traces, logs, and metrics into a Prometheus and Grafana stack, reducing system troubleshooting time.
- **Implemented CI pipelines** using Jenkins and Bitbucket to automate building, testing, and exporting new releases.

### Enghouse Systems

#### Senior Software Developer

Jan 2021 - Jan 2022 | Lisbon, Portugal (Remote, Full-time)



- **Developed a web-based, low-code IVR flow designer** using ASP.NET/C# and React/Redux, enabling non-technical users to build complex inbound logic.
- **Integrated with a Natural Language Processing (NLP) engine**, incorporating third-party services (Microsoft LUIS) to detect the intent of customers when calling the IVR.

### Altitude Software

#### AI Engineer & Researcher

May 2015 - Jan 2021 | Lisbon, Portugal (Hybrid, Full-time)



- **Designed scalable, multi-tenant Kubernetes microservices** using Python and Azure ML HyperDrive to automatically train and deploy neural network models with zero manual intervention.
- **Built an autonomous live chat recommendation engine** that, using machine learning models trained from transcripts, delivered predictive, context-aware suggestions to live agents.
- **Executed end-to-end AI research for predictive dialing**, researching and implementing Deep RL algorithms (DDPG, RDPG, Recurrent Q-Learning) in Python, and adapting a Java simulation platform (JaamSim) to rigorously test and validate performance.

## Core Skills & Technologies

### Programming Languages

- C#
- Python
- SQL
- Java

### Frameworks

- .NET / ASP.NET Core
- Entity Framework Core
- Windows Forms

### Cloud & DevOps

- Docker
- Kubernetes
- Git
- Jenkins
- Azure ML
- Google Cloud
- Apache Kafka

### Databases

- PostgreSQL
- MongoDB
- SQL Server

### Observability

- OpenTelemetry
- Prometheus
- Grafana

### Architecture

- REST APIs
- Microservices
- Event-driven architecture

## Languages

- Portuguese: Native
- English: Professional Working Proficiency (B2)

*Altitude Software*  
**Software Developer**

Jun 1997 - Apr 2015 | Lisbon, Portugal (On-site, Full-time)



- **Designed and developed uAgent Windows**, Altitude's flag-ship Windows desktop client, supporting high-concurrency voice, email, and live-chat agent interactions.
- **Architected .NET APIs and a layout editor** enabling Professional Services to build third-party integrations, embedded controls, and tailored workflow automation.
- **Implemented a high-performance call routing engine** in collaboration with Alcatel-Lucent, delivering complex, real-time call distribution logic for enterprise telephony systems.

*Megamedia*  
**Junior Web Developer**

Oct 1996 - May 1997 | Lisbon, Portugal (On-site, Full-time)



Developed web design elements and client-facing interfaces for an early enterprise online banking platform (Santander-Totta).

## Education

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*Católica Lisbon School of Business and Economics*  
Post graduation, Advanced Management Program for Executives

2001 - 2002



*Instituto Superior Técnico (University of Lisbon)*  
Licentiate Degree in Information Systems and Computer Engineering

1991 - 1996



- **Specialization:** Information Systems and Databases | **Grade: 15.3**
- **Final Project:** Designed and built a 3D Virtual Desktop application managing Windows files and applications in an interactive 3D environment.
- **Leadership:** Member of the Board of Directors, IST Student's Union (1995 - 1996).

## Training

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*Stanford Online (Andrew Ng, 2018)*  
Machine Learning



*UDEMY (Francisco Juretig, 2017)*  
Keras: Deep Learning in Python



*UDEMY (Kirill Eremenko and Hadelin de Ponteves, 2017)*  
Machine Learning A-Z: Hands-On Python & R In Data Science



*UDEMY (Jose Portilla, 2017)*  
Complete Python Bootcamp



*Nielsen Norman Group (2007)*  
Usability in Practice (Methodologies)

